Math 340 • Fall 2026 • Homework #10 • Victor Matveev

Sections 5.1-5.2: Numerical Differentiation and Numerical Integration

Problem 1 (20pts) Richardson Extrapolation for Differentiation

The centered-difference 3-point approximation for the 2nd derivative is

$$f''(x) \approx \frac{f(x-h)-2f(x)+f(x+h)}{h^2}.$$

Given that this method has **second-order accuracy**, use **Richardson extrapolation** to derive a finite difference with **fourth-order accuracy**. You do **not** have to determine the expression for the error (although this would not be hard).

Hint: apply Richardson extrapolation formula, $Q = \frac{2^n F(h)-F(2h)}{2^n-1}$, to the given approximation, $F(h) = \frac{f(x-h)-2f(x)+f(x+h)}{h^2}$, where $n = 2$ is the accuracy order of the method.

Problem 2 (50pts) Integration

Find the **error expression** and **degree of precision** of the following 2-point integration rule:

$$\int_0^h f(x)dx \approx \frac{h}{4}\left[3f\left(\frac{h}{3}\right) + f(h)\right]$$

To do this, follow these steps:

a) Evaluate the approximation for monomials, $f(x) = x^n$, with increasing *n*, until you find the smallest *n* for which the error is nonzero.

b) Using results of part "a", derive the **error term** for the rule.

c) Use the error expression in part "b" above to estimate the bound on the error in approximating $\int_0^1 e^x dx = e - 1$, and compare it to the actual numerical error of this integration rule for this integral.

Problem 3 (30pts) Romberg Integration.

Romberg integration is **Richardson extrapolation** applied sequentially (in stages), building progressively more accurate integration results starting with a less accurate method such as the Trapezoid rule.

Use this method to **fill in columns in the integral table below** for approximations to the integral $\int_0^1 \boldsymbol{dx}/(\mathbf{1}+\boldsymbol{x}^2)$, using only values in the first column (i.e. you will only be computing directly the Trapezoidal approximations in the left column):

	T_1			
	T_2	S_2		
	T_4	S_4	B_4	
	T_8	S_8	B_8	R_8
Error:	$O(h^2)$	$O(h^4)$	$O(h^6)$	$O(h^8)$

Check the accuracy of the final approximant $\boldsymbol{R_8}$, noting that $\int_0^1 \boldsymbol{dx}/(\mathbf{1}+\boldsymbol{x}^2) = \frac{\pi}{4} \approx \mathbf{0.7853981634}$.

Reminder. After completing the Trapezoidal rule values in the first column, you should compute other columns using Richardson extrapolation: $S_2 = \frac{4T_2-T_1}{3}$, $B_4 = \frac{16S_4-S_2}{15}$, etc.

Use a calculator or (better) MATLAB for all calculations, but keep at least **6-7 decimal places** after the decimal point.